Thompson's group F is nonamenable

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F.

1 The result

Thompson's group F consists of the increasing piecewise linear homeomorphisms of $[0,1]$ with finitely many pieces, dyadic rational breakpoints, and slopes in $2^{\mathbb{Z}}$. The product is composition, $hg = h \circ g$, and F is regarded as a discrete group.

A discrete group Γ is *amenable* if there is a linear functional $M : \ell^\infty(\Gamma;\mathbb{R}) \to \mathbb{R}$ with $M(1) = 1$, $M(\varphi) \geq 0$ whenever $\varphi \geq 0$, and $M(g \mapsto \varphi(hg)) = M(\varphi)$ for every $h \in \Gamma$ [19, 6]. We use only the forward implication of the Følner criterion [7]: if Γ is amenable, then for every finite $S \subset \Gamma$ and every $\varepsilon > 0$ there is a nonempty finite $A \subset \Gamma$ with

$$\frac{|hA \triangle A|}{|A|} < \varepsilon \qquad (h \in S). \tag{1.1}$$

(The formulation in [14, Theorem 1.2] bounds the sum of these ratios over $h \in S$, which gives (1.1) after shrinking ε.)

Theorem 1.1. *Thompson's group F is not amenable.*

Context. Thompson introduced F in 1965 ([11]; [5, pp. 215–216]), and Geoghegan conjectured in 1979 that it is nonamenable [5, p. 227]. F has no nonabelian free subgroup [3, Theorem 3.1] and is not elementary amenable [5, Theorem 4.10], so neither standard test decides the question; it has a classifying space with finitely many cells in each dimension [4]. Related work includes lower bounds on Følner sets [15, Theorem 1.1], a Ramsey characterization of amenability of F [14, Theorem 3.1], results on densities of Cayley subgraphs and on invariant means [8, 9], and a link with the reduced C^*-algebra of Thompson's group T [10, Theorem 5.5]. Earlier claims of a resolution are [1] (nonamenable), [18] (amenable; errors identified in [12]), and the withdrawn [13]. None of this is used below.

The one external input.

Lemma 1.2 (Benyamini–Sternfeld)**.** *There are a real Hilbert space H, a Lipschitz map $f : B \to B$ on its closed unit ball, and a constant $\delta > 0$ such that $\|f(x) - x\| \geq \delta$ for all $x \in B$.*

This is proved for every infinite-dimensional normed space in [2, Theorem, part (3), p. 439]. Appendix A gives a self-contained construction on $L^2([0,1];\mathbb{R}^2)$ with $\delta = 1/2$. The lemma is used to define the colors in Section 3 and in Step 3 of Proposition 4.2. No such map exists in finite dimensions (Brouwer), so infinite dimensionality is essential.

Proof at a glance. By (1.1), it suffices to find one finite $S \subset F$ and one $c > 0$ such that every nonempty finite $A \subset F$ has $|hA \triangle A| \geq c|A|$ for some $h \in S$. This is Proposition 4.2.

Key idea. Fix f, L, δ as in Lemma 1.2 (L a Lipschitz constant) and a large integer D. Choose D separated dyadic intervals $I_1 < \cdots < I_D$ in $(0,1)$ ("parents") and inside each I_i the scaled copies $I_i \cdot I_j$ ("descendants"). To each finite dyadic partition we attach recursively a "color" in B, so that the color seen inside a parent is f of the mean of the colors seen inside its D descendants. Evaluating on the partitions $gT^{(n)}$ ($g \in F$, $T^{(n)}$ the uniform partition of level n) gives B-valued functions X_I on F with

$$m = \frac{1}{D}\sum_{i=1}^{D} X_{I_i}, \qquad z_i = \frac{1}{D}\sum_{j=1}^{D} X_{I_i \cdot I_j}, \qquad X_{I_i} = f(z_i).$$

Elements of F can move any separated pair of these intervals affinely onto (I_1, I_2). Let S be a finite set of such elements, and η the largest relative S-boundary of A. Then the A-averages of all inner products $\langle X_I, X_J \rangle$ over separated pairs lie within η of one number α. Expanding the average of $\|z_i - m\|^2$, the α's cancel, and only a fraction $1/D$ of the terms (diagonal or nested pairs) is uncontrolled. So $z_i \approx m$ on average, hence $m = \frac{1}{D}\sum f(z_i) \approx f(m)$, contradicting $\|f(m) - m\| \geq \delta$ once D is large and η is small.

Dependencies.

- Lemma 2.1 (restriction calculus): (c) gives $X_{I_i} = f(z_i)$ (Step 2); (d) gives the transport identity (Step 1).
- Lemma 2.2: one level n works for all of the finite set $A \cup SA$.
- Lemma 2.3: existence of the elements of S.
- Lemma 4.1: translating a finite average costs at most the relative boundary (Step 1).
- Proposition 4.2: Steps 1–6 combine these with Lemma 1.2. Theorem 1.1 follows.

Section 5 records two corollaries that rely on companion papers and are not used for Theorem 1.1.

2 Dyadic partitions and affine transport

We use the standard dyadic-partition model of F [5, Section 2, Lemma 2.2]. Everything needed is proved here.

Definitions. A *basic interval* is $[k2^{-r}, (k+1)2^{-r}]$ with integers $r \geq 0$, $0 \leq k < 2^r$. A *basic partition* is a finite family of basic intervals (*cells*) covering $[0,1]$ with pairwise disjoint interiors. Its *mesh* is its largest cell length. A partition T *respects* an interval I if I is a union of cells of T. $T^{(n)}$ is the uniform partition into basic intervals of length 2^{-n}, and $gT^{(n)} = \{g(C) : C \in T^{(n)}\}$ for $g \in F$. For a basic interval I, let $s_I : [0,1] \to I$ be the increasing affine bijection, and put $I \cdot J = s_I(J)$. If T respects I, its *normalized restriction* is

$$T_I = \{s_I^{-1}(C) : C \in T,\ C \subseteq I\}.$$

Write $I < J$ if $\max I < \min J$ (so there is a positive gap). Two intervals are *separated* if $I < J$ or $J < I$. An interval is *internal* if it lies in $(0,1)$.

Lemma 2.1 (Restriction calculus)**.** *Let I, J be basic intervals and T a basic partition.*

(a) *$I \cdot J$ is basic. If T respects I, then T_I is a basic partition with as many cells as T has inside I.*

(b) *$s_{I \cdot J} = s_I \circ s_J$.*

(c) *If T respects I and $I \cdot J$, then T_I respects J and*

$$(T_I)_J = T_{I \cdot J}. \tag{2.1}$$

(d) *Let $h \in F$ map I affinely onto a basic interval I'. If $gT^{(n)}$ and $(hg)T^{(n)}$ are basic and respect I and I' respectively, then*

$$\left((hg)T^{(n)}\right)_{I'} = (gT^{(n)})_I. \tag{2.2}$$

Proof. (a) Write $I = [k2^{-r}, (k+1)2^{-r}]$, so $s_I(x) = 2^{-r}(x+k)$. If $J = [l2^{-s}, (l+1)2^{-s}]$, then $s_I(J) = [(2^s k + l)2^{-(r+s)}, (2^s k + l + 1)2^{-(r+s)}]$ is basic. Conversely, a basic $C = [l2^{-s}, (l+1)2^{-s}] \subseteq I$ has $s \geq r$, and $s_I^{-1}(C) = [(l - 2^{s-r}k)2^{-(s-r)}, (l + 1 - 2^{s-r}k)2^{-(s-r)}] \subseteq [0,1]$ is basic. Since $s_I^{-1} : I \to [0,1]$ is a bijection and the cells of T inside I cover I with disjoint interiors, T_I is a basic partition.

(b) Both sides are increasing affine bijections $[0,1] \to I \cdot J$, and there is only one.

(c) A cell C of T lies in $I \cdot J = s_I(J)$ iff $C \subseteq I$ and $s_I^{-1}(C) \subseteq J$. So the cells of T_I inside J are exactly the $s_I^{-1}(C)$ with $C \in T$, $C \subseteq I \cdot J$. Their union is $s_I^{-1}(I \cdot J) = J$, so T_I respects J. Applying s_J^{-1} and (b), $(T_I)_J = \{s_{I \cdot J}^{-1}(C) : C \in T,\ C \subseteq I \cdot J\} = T_{I \cdot J}$.

(d) $h \circ s_I$ and $s_{I'}$ are both increasing affine bijections $[0,1] \to I'$, hence equal, so $s_{I'}^{-1} \circ h = s_I^{-1}$ on I. Since h is a bijection with $h(I) = I'$, the cells of $(hg)T^{(n)}$ inside I' are exactly the $h(C)$ with C a cell of $gT^{(n)}$ inside I, and $s_{I'}^{-1}(h(C)) = s_I^{-1}(C)$. □

Lemma 2.2. *For each $g \in F$, the partitions $gT^{(n)}$ are basic for all sufficiently large n, and their meshes tend to zero as $n \to \infty$. In particular, given a finite set $E \subset F$ and a finite family $\mathcal{J}$ of basic intervals, there is an n_0 such that $gT^{(n)}$ is a basic partition respecting every member of $\mathcal{J}$ whenever $g \in E$ and $n \geq n_0$.*

Proof. **Step 1 (pieces of g).** Let $0 = x_0 < \cdots < x_N = 1$ be the breakpoints of g and 2^{q_i} its slope on $[x_i, x_{i+1}]$. Since $g(0) = 0$ and $g(x_{i+1}) = g(x_i) + 2^{q_i}(x_{i+1} - x_i)$, induction shows every $g(x_i)$ is dyadic. So $g(x) = 2^{q_i}x + b_i$ on that piece, with $b_i = g(x_i) - 2^{q_i}x_i$ dyadic.

Step 2 (fine cells have basic images). Take n so large that every $x_i \in 2^{-n}\mathbb{Z}$, and $n \geq q_i$, $b_i \in 2^{q_i - n}\mathbb{Z}$ for all i. These conditions persist as n grows. A cell $[k2^{-n}, (k+1)2^{-n}]$ then lies in one piece, and its image $[k2^{q_i - n} + b_i, (k+1)2^{q_i - n} + b_i] \subseteq [0,1]$ has length $2^{-(n - q_i)}$, $n - q_i \geq 0$, and left endpoint a multiple of that length. So it is basic. The images cover $[0,1]$ with disjoint interiors because g is a homeomorphism. The mesh is $\max_i 2^{q_i - n} \to 0$.

Step 3 (fine basic partitions respect I). Two basic intervals whose interiors meet are nested: the shorter lies in a single cell of the uniform partition at the level of the longer, and that cell must be the longer one. Let T be a basic partition of mesh less than $|I|$. Each point of the interior of I lies in a cell C whose interior meets that of I. Then C, I are nested and $|C| < |I|$, so $C \subseteq I$. Thus the cells inside I cover the interior of I, and, their union being closed, all of I.

Step 4. Steps 2–3 give a threshold for each $g \in E$, $I \in \mathcal{J}$. Let n_0 be the largest. □

Lemma 2.3. *If $I < J$ and $I' < J'$ are pairs of internal basic intervals, there is an $h \in F$ carrying I affinely onto I' and J affinely onto J'.*

Proof. The complement of $I \cup J$ consists of three gaps $[0, \min I]$, $[\max I, \min J]$, $[\max J, 1]$, of positive length (as I, J are internal and $I < J$) with dyadic endpoints. The same holds for I', J'. Each gap is a union of cells of some $T^{(n)}$, so it can be partitioned into basic intervals. For each of the three pairs of corresponding gaps, repeatedly bisect a cell on the side with fewer cells until the counts agree. A bisection produces two basic intervals and raises the count by one. This yields basic

partitions $P \ni I, J$ and $P' \ni I', J'$ with equally many cells, in which I, J and I', J' occupy the same positions from left to right. Let h map the k-th cell of P increasingly and affinely onto the k-th cell of P'. Then h is an increasing homeomorphism of $[0,1]$ with dyadic breakpoints and slopes that are ratios of powers of two, so $h \in F$. □

3 The recursive coloring

Fix H, B, f as in Lemma 1.2, and $L > 0$, $\delta > 0$ with

$$\|f(x) - f(y)\| \le L\|x - y\|, \qquad \|f(x) - x\| \ge \delta \qquad (x, y \in B). \tag{3.1}$$

Fix an integer $D \ge 2$ with

$$D > \frac{4L^2}{\delta^2}, \tag{3.2}$$

and internal basic intervals $I_1 < \cdots < I_D$ (the *parents*). For example, take $2^r > 2D$ and $I_j = [(2j-1)2^{-r}, 2j2^{-r}]$.

Colors of partitions. Define $p(T) \in B$ for every basic partition T by induction on its number of cells:

$$p(T) = \begin{cases} f\Big(\dfrac{1}{D}\displaystyle\sum_{j=1}^{D} p(T_{I_j})\Big), & \text{if } T \text{ respects every } I_j, \\ 0, & \text{otherwise.} \end{cases} \tag{3.3}$$

This is well defined. In the first case, T_{I_j} is a basic partition with as many cells as T has inside I_j (Lemma 2.1(a)). This is strictly fewer than T has, because I_j is internal and T also covers $[0,1] \setminus I_j$. So the colors $p(T_{I_j}) \in B$ are already defined. Their mean lies in the convex set B, the domain of f, and $p(T) \in f(B) \subseteq B$.

Example. With the displayed parents, $T^{(0)}$ respects no I_j, so $p(T^{(0)}) = 0$. Since $(T^{(r)})_{I_j} = T^{(0)}$ and $(T^{(2r)})_{I_j} = T^{(r)}$, we get $p(T^{(r)}) = f(0)$ and $p(T^{(2r)}) = f(f(0))$.

Colors as functions on F. Let

$$\mathcal{I} = \{I_i : 1 \le i \le D\} \cup \{I_i \cdot I_j : 1 \le i, j \le D\},$$

the parents and their *descendants*. All are internal basic intervals (Lemma 2.1(a); $I_i \cdot I_j \subseteq I_i$). Call (n, g) *admissible* if $gT^{(n)}$ is a basic partition respecting every member of $\mathcal{I}$. For $n \ge 0$, $g \in F$, $I \in \mathcal{I}$ put

$$X_I^{(n)}(g) = \begin{cases} p\big((gT^{(n)})_I\big), & \text{if } (n,g) \text{ is admissible,} \\ 0, & \text{otherwise,} \end{cases} \qquad \varphi_{I,J}^{(n)}(g) = \langle X_I^{(n)}(g), X_J^{(n)}(g) \rangle. \tag{3.4}$$

So $X_I^{(n)} : F \to B$ and $\varphi_{I,J}^{(n)} : F \to [-1,1]$ are defined on all of F.

4 A uniform boundary bound

For every ordered pair $I < J$ in $\mathcal{I}$, use Lemma 2.3 to choose $h_{I,J} \in F$ carrying I affinely onto I_1 and J affinely onto I_2. Let S be the finite set of these elements. It depends only on D and the parents, not on the set A below.

Lemma 4.1. *For nonempty finite $A \subset F$, $h \in F$ and $\varphi : F \to [-1,1]$,*

$$\Big|\frac{1}{|A|}\sum_{g\in A}\varphi(hg) - \frac{1}{|A|}\sum_{g\in A}\varphi(g)\Big| \le \frac{|hA \triangle A|}{|A|}. \tag{4.1}$$

Proof. $\sum_{g\in A}\varphi(hg) = \sum_{x\in hA}\varphi(x)$. Subtracting $\sum_{x\in A}\varphi(x)$, the terms with $x \in hA \cap A$ cancel. This leaves $\sum_{hA\setminus A}\varphi - \sum_{A\setminus hA}\varphi$, of absolute value at most $|hA \triangle A|$. □

Proposition 4.2. *For every nonempty finite set $A \subset F$,*

$$\max_{h\in S}\frac{|hA \triangle A|}{|A|} \ge \frac{\delta^2/L^2 - 4/D}{4(1-1/D)} > 0. \tag{4.2}$$

Proof. Plan. Step 1 shows that all separated pairs have almost the same averaged correlation. Steps 2–3 give a pointwise inequality $\delta^2 \le \frac{L^2}{D}\sum_i \|z_i - m\|^2$. Steps 4–5 bound the average of the right-hand side by correlations. Step 6 concludes.

Setup. Fix A and put

$$\eta = \max_{h\in S}\frac{|hA \triangle A|}{|A|}. \tag{4.3}$$

The set $A \cup SA$, $SA = \bigcup_{h\in S} hA$, is finite. By Lemma 2.2 (with $\mathcal{J} = \mathcal{I}$) there is one n with (n, g) admissible for all $g \in A \cup SA$. This n may depend on A. Fix it, drop the superscript (n), and write

$$\mathbb{E}_A\psi = \frac{1}{|A|}\sum_{g\in A}\psi(g), \qquad \alpha = \mathbb{E}_A\varphi_{I_1,I_2}.$$

Step 1 (comparing correlations). *Claim:*

$$\big|\mathbb{E}_A\langle X_I, X_J\rangle - \alpha\big| \le \eta \qquad \text{for all separated } I, J \in \mathcal{I}. \tag{4.4}$$

By symmetry of the inner product we may assume $I < J$. Let $h = h_{I,J}$ and $g \in A$. Both (n, g) and (n, hg) are admissible, since $g, hg \in A \cup SA$. So $gT^{(n)}$ is basic and respects I, J, and $(hg)T^{(n)}$ is basic and respects I_1, I_2. As h carries I onto I_1 and J onto I_2 affinely, (2.2) applied twice gives $((hg)T^{(n)})_{I_1} = (gT^{(n)})_I$ and $((hg)T^{(n)})_{I_2} = (gT^{(n)})_J$. By (3.4),

$$X_I(g) = X_{I_1}(hg), \qquad X_J(g) = X_{I_2}(hg), \qquad \varphi_{I,J}(g) = \varphi_{I_1,I_2}(hg).$$

Averaging over $g \in A$ and applying Lemma 4.1 to $\varphi = \varphi_{I_1,I_2}$ gives $|\mathbb{E}_A\varphi_{I,J} - \alpha| \le |hA \triangle A|/|A| \le \eta$.

(Admissibility of hg was arranged through the choice of n, not deduced. h need not preserve the other intervals of $\mathcal{I}$.)

The separated pairs used below are: (i) I_k, I_l with $k \ne l$; (ii) siblings $I_i \cdot I_j,\ I_i \cdot I_l$ with $j \ne l$, the images of the separated I_j, I_l under the increasing map s_{I_i}; (iii) $I_i \cdot I_j,\ I_k$ with $k \ne i$, because $I_i \cdot I_j \subseteq I_i$. Pairs $I_i \cdot I_j,\ I_i$ are nested, and nothing is claimed for them.

Step 2 (the recursion on F). For $g \in A$ put

$$m(g) = \frac{1}{D}\sum_{k=1}^{D} X_{I_k}(g), \qquad z_i(g) = \frac{1}{D}\sum_{j=1}^{D} X_{I_i\cdot I_j}(g) \quad (1 \le i \le D).$$

These are means of points of B, hence in B. *Claim:*

$$X_{I_i}(g) = f(z_i(g)) \qquad (g \in A,\ 1 \le i \le D). \tag{4.5}$$

Let $P = gT^{(n)}$. By admissibility, P respects I_i and every $I_i \cdot I_j$. By Lemma 2.1(c), P_{I_i} respects every I_j and $(P_{I_i})_{I_j} = P_{I_i \cdot I_j}$. So the first case of (3.3) applies to P_{I_i}: $X_{I_i}(g) = p(P_{I_i}) = f(\frac{1}{D}\sum_j p(P_{I_i \cdot I_j})) = f(z_i(g))$.

Step 3 (pointwise inequality). Fix $g \in A$ and omit it. By (4.5), $m = \frac{1}{D}\sum_i f(z_i)$, so $m - f(m) = \frac{1}{D}\sum_i (f(z_i) - f(m))$. Using (3.1) (displacement at $m \in B$), convexity of $\|\cdot\|^2$, and (3.1) (Lipschitz),

$$\delta^2 \le \|m - f(m)\|^2 \le \frac{1}{D}\sum_{i=1}^{D} \|f(z_i) - f(m)\|^2 \le \frac{L^2}{D}\sum_{i=1}^{D} \|z_i - m\|^2. \tag{4.6}$$

Step 4 (three second moments). Expand by bilinearity and average over A.

$\|m\|^2 = D^{-2}\sum_{k,l}\langle X_{I_k}, X_{I_l}\rangle$. The $D(D-1)$ terms with $k \neq l$ are of type (i), each with average at most $\alpha + \eta$ by (4.4). The D diagonal terms are at most 1. Hence

$$\mathbb{E}_A\|m\|^2 \le \Big(1 - \frac{1}{D}\Big)(\alpha + \eta) + \frac{1}{D}. \tag{4.7}$$

$\|z_i\|^2 = D^{-2}\sum_{j,l}\langle X_{I_i \cdot I_j}, X_{I_i \cdot I_l}\rangle$. The same count, with type (ii) pairs, gives

$$\mathbb{E}_A\|z_i\|^2 \le \Big(1 - \frac{1}{D}\Big)(\alpha + \eta) + \frac{1}{D}. \tag{4.8}$$

$\langle z_i, m\rangle = D^{-2}\sum_{j,k}\langle X_{I_i \cdot I_j}, X_{I_k}\rangle$. The $D(D-1)$ terms with $k \neq i$ are of type (iii), each with average at least $\alpha - \eta$. The D terms with $k = i$ pair a descendant with its own parent and are at least -1, as both vectors lie in B. Hence

$$\mathbb{E}_A\langle z_i, m\rangle \ge \Big(1 - \frac{1}{D}\Big)(\alpha - \eta) - \frac{1}{D}. \tag{4.9}$$

Step 5 (variance estimate). Since $\|z_i - m\|^2 = \|z_i\|^2 + \|m\|^2 - 2\langle z_i, m\rangle$, (4.7)–(4.9) give

$$\begin{aligned}\mathbb{E}_A\|z_i - m\|^2 &\le 2\Big[\Big(1 - \frac{1}{D}\Big)(\alpha + \eta) + \frac{1}{D}\Big] - 2\Big[\Big(1 - \frac{1}{D}\Big)(\alpha - \eta) - \frac{1}{D}\Big] \\ &= \frac{4}{D} + 4\Big(1 - \frac{1}{D}\Big)\eta \qquad (1 \le i \le D).\end{aligned} \tag{4.10}$$

The α terms cancel, so no information about α is needed.

Step 6 (conclusion). Average (4.6) over A and insert (4.10):

$$\delta^2 \le \frac{L^2}{D}\sum_{i=1}^{D}\mathbb{E}_A\|z_i - m\|^2 \le L^2\Big(\frac{4}{D} + 4\Big(1 - \frac{1}{D}\Big)\eta\Big).$$

Since $D \ge 2$, we may divide by $4L^2(1 - 1/D) > 0$. This gives (4.2), whose right-hand side is positive by (3.2) and independent of A. □

Proof of Theorem 1.1. Let $c > 0$ be the right-hand side of (4.2). If F were amenable, (1.1) with this S and $\varepsilon = c$ would give a nonempty finite A with $\max_{h \in S} |hA \triangle A|/|A| < c$, contradicting Proposition 4.2. □

5 Consequences

Neither corollary is used above. To avoid clashes, the source's symbols H, S (first corollary) and S, T_p (second) are renamed $\mathcal{K}, V$ and $\Sigma, \mathcal{T}_p$.

Corollary 5.1. *For every $\varepsilon > 0$, there are a separable complex Hilbert space $\mathcal{K}$ and a representation $\pi : F \to \mathrm{GL}(\mathcal{K})$ such that $\sup_{g \in F} \|\pi(g)\| \le 1 + \varepsilon$, but no bounded invertible operator V on $\mathcal{K}$ makes every $V\pi(g)V^{-1}$ unitary.*

Proof. F is countable (an element is determined by finitely many dyadic breakpoints and slope exponents) and nonamenable by Theorem 1.1. Apply the companion theorem [17, Theorem 1.1]. □

For a finite symmetric generating set $\Sigma \subset F \setminus \{\mathrm{id}\}$, let G_Σ be the simple undirected Cayley graph with edges $\{x, xs\}$. In Bernoulli bond percolation each edge is independently open with probability p. Let p_c be the threshold for existence of an infinite open cluster and p_u the threshold for almost-sure uniqueness of the infinite open cluster. The connection kernel $\mathcal{T}_p(x, y) = \mathbb{P}_p(x \leftrightarrow y)$ acts on counting-measure $\ell^2(F)$, initially on finitely supported functions by summation over y. Put $p_{2\to 2} = \sup\{p \in [0, 1] : \mathcal{T}_p \text{ is bounded on } \ell^2(F)\}$.

Corollary 5.2. *For every such generating set Σ,*

$$\|\mathcal{T}_{p_c}\|_{2\to 2} < \infty, \qquad p_c < p_{2\to 2} \le p_u.$$

For each fixed $p \in (p_c, p_u)$, there are almost surely infinitely many infinite open clusters. Moreover, there are deterministic $p_c < p_1 < p_2 < 1$, depending on Σ, such that in the coupling with independent uniform $[0, 1]$ edge labels U_e, opening e when $U_e \le p$, there are almost surely infinitely many infinite open clusters simultaneously for every $p \in [p_1, p_2]$.

Proof. Apply the companion result [16, Corollary 1.2] to F, nonamenable by Theorem 1.1. □

Unverified step. Both corollaries are one-line applications of companion theorems whose statements are not reproduced in the source, and I could not check them. As far as I know, the general statements they would need (such representations for every countable nonamenable group; these percolation conclusions for every Cayley graph of a nonamenable group) are not established in the earlier literature. The corollaries are therefore only as reliable as those two references.

A An explicit map with positive displacement

This appendix proves Lemma 1.2 with $\delta = 1/2$, giving the Hilbert-space instance of [2] directly. Its notation is local: $r, d, K, c, t, \ldots$ are unrelated to Sections 2–4. For $\xi = (\xi_1, \xi_2) \in \mathbb{R}^2$ write $\xi^\perp = (-\xi_2, \xi_1)$ (the source's $J\xi$).

Proposition A.1. *Let $H = L^2([0, 1]; \mathbb{R}^2)$, a real Hilbert space, and $B = \{x \in H : \|x\| \le 1\}$. There is a globally Lipschitz map $f : B \to B$ with*

$$\|f(x)\| = 1, \qquad \|f(x) - x\| \ge \frac{1}{2} \qquad (x \in B).$$

Plan. We build a Lipschitz $G : B \to H$ with $\|G\|$ bounded below and $G(x) = x$ for $\|x\| \geq 1/2$. Then $f = -G/\|G\|$ works. G is the identity outside a thin tube around a curve γ through 0. The curve is a straight ray to infinity for negative parameters and stays in a small ball for positive ones, without ever approaching itself (Lemma A.2). Inside the tube, write the nearest curve point as $\gamma(t) = a(t)v(t)$ with $v(t)$ a unit vector. Near the curve, G replaces the coefficient $a(t)$ of $v(t)$ by $\min\{a(t), -1/8\}$ and rotates the residual $x - \gamma(t)$, with cutoffs that return G to the identity at the edge of the tube; see (A.5). In this way nothing lands at 0.

The curve. Define

$$a(t) = \begin{cases} t/8, & t \leq 2, \\ 5/16 - (3-t)^2/16, & 2 < t < 3, \\ 5/16, & t \geq 3, \end{cases} \qquad \theta(t) = \begin{cases} 0, & t \leq 1, \\ (t-1)^2/2, & 1 < t < 2, \\ t - 3/2, & t \geq 2, \end{cases}$$

and, for $w \in [0, 1]$,

$$v(t)(w) = (\cos(\theta(t)w), \sin(\theta(t)w)), \qquad \gamma(t) = a(t)v(t).$$

Direct inspection (values and derivatives agree at $t = 1, 2, 3$) gives:

(F1) a is C^1, $0 \leq a' \leq 1/8$, a' is Lipschitz, $a \leq 5/16$, $a' = 1/8$ on $(-\infty, 2]$, $a \geq 1/8$ on $[1, \infty)$, $a \geq 1/4$ on $[2, \infty)$.

(F2) θ is C^1, $0 \leq \theta' \leq 1$, θ' is Lipschitz, $\theta = 0$ on $(-\infty, 1]$, θ is strictly increasing on $[1, \infty)$, $\theta' = 1$ on $[2, \infty)$.

(F3) $\|v(t)\| = 1$, and $\langle v(s), v(t)\rangle = \int_0^1 \cos((\theta(t) - \theta(s))w)\, dw = \operatorname{sinc}(\theta(t) - \theta(s))$, where $\operatorname{sinc}(q) = \sin(q)/q$, $\operatorname{sinc}(0) = 1$.

So $\gamma(0) = 0$, $\|\gamma(t)\| = |a(t)|$, and for $t \leq 1$ the curve is the line $t \mapsto (t/8)v(1)$.

Lemma A.2. *The curve γ is injective, has a bounded Lipschitz derivative, and*

$$c := \inf_{t \in \mathbb{R}} \|\gamma'(t)\| = \frac{1}{8}.$$

Its unit tangent $u(t) = \gamma'(t)/\|\gamma'(t)\|$ is Lipschitz, $\langle u(t), v(t)\rangle \geq 0$, and $u(t) = v(t)$ for $t \leq 1$. There is a radius $0 < \rho < 1/32$ such that every point in

$$U = \{x \in H : \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho\}$$

has a unique nearest point $\gamma(t(x))$. Writing $e(x) = x - \gamma(t(x))$, we have $e(x) \perp u(t(x))$, and

$$|t(x) - t(y)| \leq 16\|x - y\| \quad \text{if } x, y \in U \text{ and } \|x - y\| < \rho. \tag{A.1}$$

Proof. **Step 1 (derivative).** Since θ is C^1 with Lipschitz derivative and $0 \leq w \leq 1$, the difference quotients of $t \mapsto v(t)(w)$ converge uniformly in w, hence in H. Thus

$$\gamma'(t)(w) = a'(t)v(t)(w) + a(t)\theta'(t)\, w\, v(t)(w)^{\perp}, \qquad \|\gamma'(t)\|^2 = a'(t)^2 + \frac{a(t)^2\theta'(t)^2}{3},$$

using $v \perp v^{\perp}$ pointwise and $\int_0^1 w^2 dw = 1/3$. For $t \leq 2$ the first term is $1/64$, with equality in total for $t \leq 1$ where $\theta' = 0$. For $t \geq 2$ the second term is at least $(1/16)/3 > 1/64$ by (F1), (F2). So $c = 1/8$.

Step 2 (regularity). The coefficients a' and $a\theta'$ are bounded and Lipschitz: $a\theta'$ vanishes on $(-\infty, 1]$, and on $[1, \infty)$ both factors are bounded and Lipschitz. The vectors $v(t)$ and $w\,v(t)^\perp$ have norm at most 1 and are 1-Lipschitz in t, since $|\theta(t) - \theta(s)| \le |t - s|$. Hence γ' is bounded and Lipschitz. As $\|\gamma'\| \ge 1/8$, u is Lipschitz. Moreover $\langle u, v\rangle = a'/\|\gamma'\| \ge 0$, and for $t \le 1$, $\gamma' = v/8$, so $u = v$.

Step 3 (injectivity). Let $s < t$. If $t \le 1$, then $\gamma(s) \ne \gamma(t)$ because a is strictly increasing there and v is constant. If $t > 1$, then $\theta(t) \ne \theta(s)$ by (F2). So $|\langle v(s), v(t)\rangle| < 1$ by (F3) and $|\operatorname{sinc}(q)| < 1$ for $q \ne 0$. The unit vectors $v(s), v(t)$ are thus not collinear, and $a(t) > 0$, so $a(t)v(t) \ne a(s)v(s)$.

Step 4 (uniform separation). *Claim:*

$$\sigma(d) := \inf_{|t-s|\ge d} \|\gamma(t) - \gamma(s)\| > 0 \qquad (d > 0). \tag{A.2}$$

Otherwise choose $s_n < t_n$ with $t_n - s_n \ge d$ and $\|\gamma(t_n) - \gamma(s_n)\| \to 0$, and pass to a subsequence with $s_n \to s$, $t_n \to t$ in $[-\infty, \infty]$. Each case is contradictory:

- s, t finite: $t - s \ge d$ and $\gamma(s) = \gamma(t)$, against Step 3.
- $s = t = -\infty$: eventually $s_n, t_n \le 1$ and the distance is $(t_n - s_n)/8 \ge d/8$.
- $s = -\infty < t$: $\|\gamma(s_n)\| = |a(s_n)| \to \infty$ while $\|\gamma(t_n)\|$ stays bounded.
- s finite, $t = +\infty$: $\theta(t_n) - \theta(s_n) \to \infty$, so $\langle v(s_n), v(t_n)\rangle \to 0$ and $\|\gamma(t_n) - \gamma(s_n)\|^2 \to a(s)^2 + (5/16)^2 > 0$.
- $s = t = +\infty$: eventually $s_n, t_n \ge 3$, and $\|\gamma(t_n) - \gamma(s_n)\|^2 = 2(5/16)^2(1 - \operatorname{sinc}(t_n - s_n))$. This is bounded below because $\sup_{q\ge d} \operatorname{sinc}(q) < 1$ (sinc is continuous, less than 1 off 0, and $|\operatorname{sinc}(q)| \le 1/q$).

Step 5 (tangent estimate, choice of ρ). Let $K = \operatorname{Lip}(\gamma')$, $K_u = \operatorname{Lip}(u)$, and fix $d > 0$ with $Kd \le c/2$. Since $\gamma(t) - \gamma(s) - (t - s)\gamma'(s) = \int_s^t (\gamma'(\tau) - \gamma'(s))d\tau$ has norm at most $\frac{K}{2}|t - s|^2$, and $\langle \gamma'(s), u(s)\rangle = \|\gamma'(s)\| \ge c$,

$$|\langle \gamma(t) - \gamma(s), u(s)\rangle| \ge c|t - s| - \frac{K}{2}|t - s|^2 \ge \frac{3c}{4}|t - s| \qquad (|t - s| \le d). \tag{A.3}$$

Choose $\rho > 0$ with

$$\rho < \frac{1}{32}, \qquad 4\rho < \sigma(d), \qquad \rho K_u < \frac{c}{4}. \tag{A.4}$$

Step 6 (nearest points exist). Let $x \in U$ and let $\gamma(t_n)$ be a minimizing sequence for $\operatorname{dist}(x, \gamma(\mathbb{R})) < \rho$. Late terms satisfy $\|x - \gamma(t_n)\| < 2\rho$, so they are pairwise closer than $4\rho < \sigma(d)$. By (A.2) their parameters differ by less than d. The late t_n thus lie in a compact interval, and a convergent subsequence attains the minimum. At any minimizing t, the derivative of $\|x - \gamma(t)\|^2$ vanishes, i.e. $(x - \gamma(t)) \perp \gamma'(t)$.

Step 7 (parameter estimate, uniqueness). Let $x, y \in U$, $\|x - y\| < \rho$, with any minimizing parameters t, s, and write $x = \gamma(t) + e_t$, $y = \gamma(s) + e_s$, where $\|e_t\|, \|e_s\| < \rho$. Then $\|\gamma(t) - \gamma(s)\| < 3\rho < \sigma(d)$, so $|t - s| < d$. Since $e_s \perp u(s)$ and $e_t \perp u(t)$,

$$\|x - y\| \ge |\langle x - y, u(s)\rangle| \ge |\langle \gamma(t) - \gamma(s), u(s)\rangle| - |\langle e_t, u(s) - u(t)\rangle| \ge \Big(\frac{3c}{4} - \rho K_u\Big)|t - s| \ge \frac{c}{2}|t - s|,$$

by (A.3) and (A.4). With $y = x$ this gives $t = s$ (uniqueness). In general it is (A.1), as $c/2 = 1/16$. □

The next lemma turns a residual perpendicular to the tangent $u(t)$ into one perpendicular to $v(t)$. It is used in the lower bound for $\|G\|$.

Lemma A.3. *There is a family of orthogonal operators $R_t : H \to H$, Lipschitz in operator norm, such that $R_t u(t) = v(t)$ and $R_t = \mathrm{Id}$ for $t \le 1$.*

Proof. Suppress t, put $\kappa = \langle u, v\rangle \ge 0$, and define

$$Ah = v\langle u, h\rangle - u\langle v, h\rangle, \qquad R = \mathrm{Id} + A + \frac{A^2}{1+\kappa}.$$

A is skew-adjoint: $\langle Ah, k\rangle = \langle v, k\rangle\langle u, h\rangle - \langle u, k\rangle\langle v, h\rangle = -\langle h, Ak\rangle$. Also $Au = v - \kappa u$, $Av = \kappa v - u$, whence $A^2 u = -(1-\kappa^2)u$ and $A^2 v = -(1-\kappa^2)v$.

If u, v are independent, then on their span $R = \kappa\,\mathrm{Id} + A$. So $Ru = v$ and $R^*R = (\kappa - A)(\kappa + A) = \kappa^2 - A^2 = \mathrm{Id}$ there. On the orthogonal complement $A = 0$ and $R = \mathrm{Id}$. Hence R is orthogonal. If u, v are dependent unit vectors, then $u = v$ because $\kappa \ge 0$, so $A = 0$ and $R = \mathrm{Id}$.

A_t is a sum of products of the bounded Lipschitz functions u, v, and $1/(1+\kappa)$ is Lipschitz since $1 + \kappa \ge 1$. So $t \mapsto R_t$ is Lipschitz. For $t \le 1$, $u = v$ (Lemma A.2) and $R_t = \mathrm{Id}$. □

Proof of Proposition A.1. Fix ρ, U, $t(\cdot)$, $e(\cdot)$ from Lemma A.2 and R_t from Lemma A.3. Set

$$b(t) = \min\{a(t), -1/8\}, \qquad q(t) = b(t) - a(t) = -\max\{a(t) + 1/8, 0\},$$

so $|q| \le 7/16$ (as $a \le 5/16$), $\mathrm{Lip}(q) \le 1/8$, and $q(t) = 0$ when $a(t) \le -1/8$. Let $\beta, \chi : [0, \infty) \to [0, 1]$ be

$$\beta(r) = \begin{cases} 1, & r \le \rho/4, \\ 2 - 4r/\rho, & \rho/4 < r < \rho/2, \\ 0, & r \ge \rho/2, \end{cases} \qquad \chi(r) = \begin{cases} 1, & r \le \rho/2, \\ 2 - 2r/\rho, & \rho/2 < r < \rho, \\ 0, & r \ge \rho. \end{cases}$$

Define $G(x) = x$ for $x \in B \setminus U$. For $x \in B \cap U$, with $t = t(x)$, $e = e(x)$, $r = \|e\| = \mathrm{dist}(x, \gamma(\mathbb{R}))$, define

$$\begin{aligned} G(x) &= \gamma(t) + \beta(r)q(t)v(t) + \big((1 - \chi(r))\mathrm{Id} + \chi(r)R_t\big)e \\ &= x + \beta(r)q(t)v(t) + \chi(r)(R_t - \mathrm{Id})e. \end{aligned} \tag{A.5}$$

Step 1 (G is Lipschitz). Consider four kinds of pairs $x, y \in B$.

Both outside U: G is the identity.

Both in U, $\|x - y\| < \rho$: by (A.1), $t(\cdot)$ changes by at most $16\|x - y\|$. With $M = \sup\|\gamma'\|$, $\|e(x) - e(y)\| \le (1 + 16M)\|x - y\|$. The distance r is 1-Lipschitz, and $\beta, \chi, q, v, t \mapsto R_t$ are Lipschitz. So each factor in the second line of (A.5) changes by at most a constant times $\|x - y\|$. Each is bounded ($\beta, \chi \le 1$, $|q| \le 7/16$, $\|v\| = 1$, $\|R_t - \mathrm{Id}\| \le 2$, $\|e\| < \rho$). Products of such factors satisfy the same kind of estimate, with one constant for all these pairs.

$\|x - y\| \ge \rho$: by the second line of (A.5), $\|G(x)\| \le 1 + 7/16 + 2\rho < 3/2$ on B, so $\|G(x) - G(y)\| \le 3 \le (3/\rho)\|x - y\|$.

$x \in U$, $y \notin U$, $\|x - y\| < \rho$: both cutoffs satisfy $\beta(r) \le \chi(r) \le 2(\rho - r)/\rho$, so $\|G(x) - x\| \le \frac{2(\rho - r)}{\rho}(\frac{7}{16} + 2\rho) = (\frac{7}{8\rho} + 4)(\rho - r)$. Since the distance to the curve is 1-Lipschitz and is at least ρ at y, $\rho - r \le \|x - y\|$. With $G(y) = y$ this bounds $\|G(x) - G(y)\| \le \|G(x) - x\| + \|x - y\|$.

Step 2 (lower bound). *Claim:*

$$\|G(x)\| \ge \rho/4 \qquad (x \in B). \tag{A.6}$$

Outside U: $0 = \gamma(0)$ lies on the curve, so $\|G(x)\| = \|x\| \ge \rho$.

Inside U, $r \le \rho/2$: $\chi = 1$, so $G(x) = (a(t) + \beta(r)q(t))v(t) + R_t e$. Since $e \perp u(t)$ and R_t is orthogonal with $R_t u(t) = v(t)$, we have $R_t e \perp v(t)$ and $\|R_t e\| = r$. If $r \le \rho/4$, then $\beta = 1$ and

the coefficient of $v(t)$ is $b(t) \le -1/8$, so $\|G(x)\| \ge 1/8 > \rho/4$. If $\rho/4 \le r \le \rho/2$, the perpendicular component alone has norm $r \ge \rho/4$.

Inside U, $\rho/2 \le r < \rho$: $\beta = 0$, and the last term in the first line of (A.5) has norm at most $r < \rho$. If $t > 1$, then $\|\gamma(t)\| = a(t) \ge 1/8$ and $\|G(x)\| \ge 1/8 - \rho > \rho/4$. If $t \le 1$, then $R_t = \mathrm{Id}$ and $e \perp u(t) = v(t)$, so $G(x) = a(t)v(t) + e$ has norm at least $r \ge \rho/2$.

Step 3 ($G(x) = x$ when $\|x\| \ge 1/2$). This holds by definition outside U. Inside U, $|a(t)| = \|\gamma(t)\| \ge \|x\| - r > 1/2 - \rho > 5/16$. Since $a \le 5/16$ everywhere, this forces $a(t) < -5/16 < -1/8$, hence $t \le 1$. Thus $q(t) = 0$ and $R_t = \mathrm{Id}$, and (A.5) gives $G(x) = x$.

Step 4 (the map f). Define $f(x) = -G(x)/\|G(x)\|$. On vectors of norm at least $\rho/4$, normalization is Lipschitz with constant at most $8/\rho$. So f is Lipschitz by Steps 1–2, and $\|f(x)\| = 1$. If $\|x\| < 1/2$, then $\|f(x) - x\| \ge 1 - \|x\| > 1/2$. Otherwise $G(x) = x$ by Step 3, so $f(x) = -x/\|x\|$ and $\|f(x) - x\| = 1 + \|x\| \ge 3/2$. □

Verification checklist

1. **Transport identity** (Proposition 4.2, Step 1; Lemma 2.1(d)): $X_I(g) = X_{I_1}(h_{I,J}g)$ needs admissibility at both g and $h_{I,J}g$. Check that n is chosen for all of $A \cup SA$ and that S is fixed before A.
2. **Which pairs are separated** (list (i)–(iii) in Step 1) and the counts $D(D-1)$ versus D in Step 4. The nested pairs $(I_i \cdot I_j, I_i)$ are the only uncontrolled mixed terms.
3. **Recursion on F** (Step 2): $X_{I_i} = f(z_i)$ uses Lemma 2.1(c) and the first case of (3.3). Check also that p is well defined (Section 3).
4. **Cancellation of α** (Step 5) and the final arithmetic (Step 6) with $D > 4L^2/\delta^2$.
5. **Pair transport in F** (Lemma 2.3) and **fine images are basic** (Lemma 2.2, Steps 2–3).
6. **Existence of f** (Appendix A): the uniform separation (A.2) (Lemma A.2, Step 4), the lower bound (A.6), and $G = \mathrm{id}$ on $\|x\| \ge 1/2$ (Steps 2–3 of the proof of Proposition A.1). Alternatively, accept Lemma 1.2 from [2].
7. **Corollaries** (Section 5) rest on companion papers not checked here. Theorem 1.1 does not depend on them.

References

[1] Azer Akhmedov. Non-amenability of R. Thompson's group F. arXiv:1310.4395v3, 2021.

[2] Y. Benyamini and Y. Sternfeld. Spheres in infinite-dimensional normed spaces are Lipschitz contractible. *Proceedings of the American Mathematical Society*, 88(3):439–445, July 1983. doi:10.1090/S0002-9939-1983-0699410-7.

[3] Matthew G. Brin and Craig C. Squier. Groups of piecewise linear homeomorphisms of the real line. *Inventiones Mathematicae*, 79(3):485–498, 1985. doi:10.1007/BF01388519.

[4] Kenneth S. Brown and Ross Geoghegan. An infinite-dimensional torsion-free FP_∞ group. *Inventiones Mathematicae*, 77(2):367–381, 1984. doi:10.1007/BF01388451.

[5] J. W. Cannon, W. J. Floyd, and W. R. Parry. Introductory notes on Richard Thompson's groups. *L'Enseignement Mathématique (2)*, 42(3–4):215–256, 1996. Primary article scan.

[6] Mahlon M. Day. Amenable semigroups. *Illinois Journal of Mathematics*, 1(4):509–544, 1957. doi:10.1215/ijm/1255380675.

[7] Erling Følner. On groups with full Banach mean value. *Mathematica Scandinavica*, 3:243–254, 1955. doi:10.7146/math.scand.a-10442.

[8] Victor Guba. Cayley graphs of R. Thompson's group F: New estimates for the density. *Journal of Combinatorial Algebra*, 9(1/2):145–168, 2025. doi:10.4171/JCA/110.

[9] Victor Guba. On zero-measured subsets of Thompson's group F. *Journal of Algebra*, 692:106–122, April 2026. doi:10.1016/j.jalgebra.2025.12.006.

[10] Uffe Haagerup and Kristian Knudsen Olesen. Non-inner amenability of the Thompson groups T and V. *Journal of Functional Analysis*, 272(11):4838–4852, 2017. doi:10.1016/j.jfa.2017.02.003.

[11] Ralph McKenzie and Richard J. Thompson. An elementary construction of unsolvable word problems in group theory. In W. W. Boone, F. B. Cannonito, and R. C. Lyndon, editors, *Word Problems: Decision Problems and the Burnside Problem in Group Theory*, volume 71 of *Studies in Logic and the Foundations of Mathematics*, pages 457–478. North-Holland, Amsterdam, 1973. doi:10.1016/S0049-237X(08)71914-1.

[12] Justin Tatch Moore. A note on Shavgulidze's papers concerning the amenability problem for Thompson's group F. arXiv:1102.0747v2, 2011.

[13] Justin Tatch Moore. Nonassociative Ramsey theory and the amenability of Thompson's group. arXiv:1209.2063v4, 2012. Withdrawn.

[14] Justin Tatch Moore. Amenability and Ramsey theory. *Fundamenta Mathematicae*, 220(3):263–280, 2013. doi:10.4064/fm220-3-6.

[15] Justin Tatch Moore. Fast growth in the Følner function for Thompson's group F. *Groups, Geometry, and Dynamics*, 7(3):633–651, 2013. doi:10.4171/GGD/201.

[16] OpenAI. Nonuniqueness of percolation on nonamenable quasi-transitive graphs. OpenAI Math Release preprint OAI:Nonuniqueness-of-percolation-on-nonamenable-quasi-transitive-graphs-September-24-2026, 2026. Corollary 1.2.

[17] OpenAI. Unitarizability implies amenability for discrete groups. OpenAI Math Release preprint OAI:Unitarizability-Implies-Amenability-for-Countable-Groups-September-23-2026, 2026. Theorem 1.1.

[18] E. T. Shavgulidze. About amenability of subgroups of the group of diffeomorphisms of the interval. arXiv:0906.0107v1, 2009.

[19] John von Neumann. Zur allgemeinen Theorie des Masses. *Fundamenta Mathematicae*, 13:73–116, 1929. doi:10.4064/fm-13-1-73-116.